

■ COMPLETE C++ REFERENCE CHEAT SHEET

A complete reference guide to all major C++ programming symbols, terms, and concepts — perfect for beginners and students to quickly recall everything at a glance.

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// ■ COMPLETE C++ REFERENCE CHEAT SHEET
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// Created for beginners to understand every major symbol, term & concept in C++

#include <iostream>
#include <string>
#include <cmath>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
    cout << "Hello World!" << endl;
    return 0;
}

// ----- DATA TYPES -----
// int, float, double, char, string, bool

// ----- OPERATORS -----
// +, -, *, /, %, =, ==, !=, >, <, >=, <=, &&, ||, !, ++, --, &, |, ^, ~, <<, >>
// << → Stream Insertion Operator (cout)
// >> → Stream Extraction Operator (cin)
// :: → Scope Resolution Operator
// . → Member Access Operator
// -> → Pointer Access Operator
// ? : → Ternary Operator

// ----- CONTROL STATEMENTS -----
// if, else if, else, switch, for, while, do-while, break, continue

// ----- FUNCTIONS -----
// void greet() { ... }
// int add(int a, int b) { return a + b; }
// Function overloading and recursion supported

// ----- POINTERS -----
// int *ptr = &x; → Pointer to variable x
// *ptr → Dereference (access value)

// ----- CLASSES & OBJECTS -----
// class MyClass { public: void show(); };

// ----- INHERITANCE -----
// class Derived : public Base { };

// ----- POLYMORPHISM -----
// virtual void show();

// ----- FILE HANDLING -----
// ofstream fout("data.txt"); fout << "Hello"; fout.close();
// ifstream fin("data.txt"); getline(fin, line);

// ----- EXCEPTION HANDLING -----
// try { ... } catch (ExceptionType e) { ... }

// ----- STL CONTAINERS -----
```

```
// vector<int> v = {1,2,3}; v.push_back(4);
// map<string, int> m; m["key"] = 10;

// ----- ESCAPE SEQUENCES -----
// \n → New Line, \t → Tab, \\ → Backslash, \" → Quote

// ----- COMMENTS -----
// // Single-line, /* Multi-line */

// ----- MEMORY MANAGEMENT -----
// new, delete, new[], delete[]

// ----- END -----
```